

iamneo: Equipping thousands of students in Indian cities to succeed in the working world

Tell us your challenge. We're here to help.

Contact us (/contact)

iamneo leverages Cloud's scalability to stabilize its platform, grow its product, help student development at universities and colleges to meet the demand of the industry.

iamneo

About iamneo

Founded in 2016, [iamneo](https://iamneo.ai/) (<https://iamneo.ai/>) is a bootstrapped education technology startup based out of Coimbatore, India. It was built in response to the lack of tools for the teaching of computer science in universities in Indian cities. The company provides a platform for developer upskilling specifically for universities and enterprises with a focus on deep analytics.

Industries: Education

Google Cloud results

- 10x increase in dynamic user loads handled by higher application scalability
- Performance and response times improved from 8 to 2 seconds
- 75% decrease in monitoring and manual scaling
- 50% more visibility into infrastructure scaling

When four friends in Coimbatore noticed a lack of technology tools science students in local universities, there was a problem, one that not only affected them but also the community.

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Location: India

Cloud Functions

searche°

About Searche

Searche is a niche cloud consulting business with futuristic tech in its DNA, focused on "realizing the Next in the Now" for its clients. Specializing in cloud data engineering, AI/ML, and advanced cloud InfraTech, such as Anthos and Kubernetes, Searche is one of the top partners for Google Cloud globally, with 3,000+ clients successfully moved to the cloud.

Google Cloud (<https://cloud.google.com/>)

App Engine (<https://cloud.google.com/appengine/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Build (<https://cloud.google.com/build/>)

Cloud DNS (<https://cloud.google.com/dns/>)

Cloud Functions
(<https://cloud.google.com/functions/>)

Logging (<https://cloud.google.com/logging/>)

Cloud Run (<https://cloud.google.com/run/>)

Cloud SQL (<https://cloud.google.com/sql/>)

business with futuristic tech in its DNA,

The four, Aasif Iqbal J, Arumugh Suresh Shanmugam, and Senthil, are preparing for an exam preparation and proctoring platform for computer science graduates to meet the demands of the industry after getting into [iamneo](https://iamneo.ai) (https://iamneo.ai) in a platform for developer upskilling.

Cloud Storage (<https://cloud.google.com/storage/>)
for its clients. Specializing in cloud data

Compute Engine

(<https://cloud.google.com/compute/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Networking Products

(<https://cloud.google.com/products/networking>)

"There has always been a gap between what the industry expects and the skills that graduates bring. Universities still have a conventional mindset focused on building physical infrastructure rather than investing in technology that can help their faculty tailor a curriculum in anticipation of what the industry needs," Iqbal says.

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Vertex AI (<https://cloud.google.com/vertex-ai/>)

Cloud Security Command Center

(<https://cloud.google.com/security-command-center/>)

As a result, developers and computer engineers in their first jobs out of university often need up to 180 days of probation and training before they can deliver billable hours, says Divya Ajitkumar, iamneo's Head of Strategic Initiatives. That's where iamneo's education technology platform can help.

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Overcoming scalability challenges

Today, iamneo helps an average of 40,000 students daily to produce live GitHub-based profile resumes to demonstrate their know-how to prospective employers and ready their skills for the real world.

However, getting to this scale has been fraught with challenges.

In 2012, iamneo's earliest platform prototypes repeatedly failed due to the startup's monolith architecture that provided scalability constraints. The team's third attempt in 2016 saw them prioritize a cloud architecture over the incumbent monolith one, and they also ventured to build an in-house team of engineers proficient in Node.js, an open-source server environment.

But, again, their ambitions were hampered by the inability to find graduates with the right skill sets, which was the very problem they were looking to solve, and iamneo's incumbent cloud platform became too costly as compute costs were rising dramatically with demand.

Then, a \$100,000 [Google for Startups Cloud Program](https://cloud.google.com/startup) (<https://cloud.google.com/startup>) grant opened the door for iamneo's engineers to explore [Google Cloud](https://cloud.google.com) (<https://cloud.google.com>) and move several workloads over.

iamneo's founders were also at a point where they wanted to consolidate billing to a monthly cycle. When they were introduced to [Searce](https://www.searce.com/) (<https://www.searce.com/>) by Google, they gained more billing transparency, ease of payment, and superior technical support.

"This helped massively because we were bootstrapped and were facing a cash crunch. After getting that helping hand from Google and Searce,

we started understanding Google Cloud's power and potential, encouraging us to move more workloads over," Shanmugam explains.

The team has also introduced more personalization into each student's learning experience using BigQuery (<https://cloud.google.com/bigquery>) through custom reports and business intelligence integration.

With BigQuery, iamneo's data science team can understand studying patterns and analyze how students use its platform to devise better algorithms that can correct their learning journey or reinforce areas where they're weak.

As a result, students don't feel overwhelmed, and they can experience a steady learning pace while being constantly nudged to retain what they've learned previously.

For long-term growth, iamneo also needed to ensure that workloads could be scaled on demand in a cost-effective manner. While the platform routinely serves about 40,000 students daily, this can surge to 100,000 users when they need to prepare for exams. With Google Cloud, iamneo realized a 15% improvement in cost management.

"Most students use our platform late at night, so we need round-the-clock autoscaling without downtime. Our immediate strategy was to leverage Compute Engine's Managed Instance Groups to easily deploy virtual machine (VM) instances or new code, and then slowly shut down other instances when new code is introduced."

—Suresh Shanmugam,
Co-Founder, iamneo

An agile, serverless infrastructure for challenging use cases to help developers upskill

Next, the team turned to [Cloud SQL](https://cloud.google.com/sql) (<https://cloud.google.com/sql>), [Compute Engine](https://cloud.google.com/compute) (<https://cloud.google.com/compute>), and [Cloud Functions](https://cloud.google.com/functions) (<https://cloud.google.com/functions>) to expand service delivery at scale.

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The platform also needed to handle about 400 APIs, which iamneo's engineers solved for by moving those services into Cloud Functions with [Pub/Sub](https://cloud.google.com/pubsub) (<https://cloud.google.com/pubsub>) queues.

They also realized that maintaining iamneo's codebase and 45 compilers in VMs would be unsustainable, so they have started utilizing microservices. Here's where Searce stepped in again to help, and they introduced iamneo to [Cloud Build](https://cloud.google.com/build) (<https://cloud.google.com/build>) for their CI/CD purposes and talked the team through adopting [Cloud Run](https://cloud.google.com/run) (<https://cloud.google.com/run>) to deploy its

codebase. As a result, iamneo's engineers have seen response time improvements from the previous eight seconds to under two seconds.

The move to serverless architecture has reduced monitoring and manual scaling by 75% and increased infrastructure visibility by 50%. It has also freed up iamneo's engineering team to pursue business innovations rather than having to worry about managing their infrastructure.

One new use case arising from this newfound ability for innovation is to serve corporate clients and help enterprise developers upskill. These enterprise developers need sandboxed environments with the freedom for trial and error and the intelligence to auto-validate their code.

"We used Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine>) to run Kubernetes clusters to create sandbox environments for students, which we provide in neocoder, our sandbox product," Shanmugam says.

To ease its corporate clients' security concerns, the iamneo team piloted Security Command Center (<https://cloud.google.com/security-command-center>) with Searce to better secure the Google Cloud infrastructure, which their platform was built upon. This is part and parcel of iamneo's mission to achieve SOC 2 compliance as it aims to expand its reach to global clients.

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Expanding reach to developers from around the world

As iamneo looks to onboard more global corporate clients onto its platform, they are aware of these clients' need to hire larger scales of engineers. They are developing a product that can help companies' learning and development teams to better engage with fresh engineering talent.

Shanmugam shares that student developers spend an average of 20 hours per week on their platform and are committed to practicing problems, going through hackathons, and exploring sandboxes to quickly understand the quality of their code.

"Corporates need large-scale visibility over every employee's progress. Moving forward, we are looking to continue leveraging the capabilities of Google Cloud to make our platform extremely usable and improve the personalized learning experience for our students. We are also focusing on extending auto evaluation to all aspects of our platform," Shanmugam concludes.